

The Light Bulb Your Grandmother Bought

By Roleigh Martin

Your grandmother bought a light bulb without thinking about it. She screwed it in, and it threw a warm, even glow across the kitchen. That glow was not an accident of taste. It was the physics of a hot metal filament, and it happened to match, more closely than anything you can easily buy new today, the light under which the human body spent its whole history.

Three bulb technologies have shared American sockets in living memory: the incandescent, the compact fluorescent (CFL) and the light-emitting diode (LED). Each replaced the last in the name of saving energy. Each also changed the light itself—its color, its rhythm, the invisible wavelengths it carries and the ones it leaves out. Those changes reach the eye, the skin and the internal clock that governs sleep.

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The energy argument is real. An incandescent bulb spends most of its power making heat, not light, which is why it feels warm to the hand and why the newer bulbs cost less to run. The debate has almost always been about the electric bill, almost never about what the light does to the person sitting under it. That second question is the one a health journal should ask. This is what the peer-reviewed science says, one technology at a time—what each bulb does well, what it does badly, and what the light we retired was quietly doing for us that we never noticed until it was gone.

Consider how fast this happened. The incandescent bulb lit American homes for more than a century, essentially unchanged since Edison's day. The compact fluorescent arrived slowly—the spiral design dates to 1976, and the first bulb reached the market around 1980—but for decades, it stayed a niche product; as late as 2000, it constituted under 1 percent of American bulb sales. By 2007, the CFL had climbed past a fifth of bulb sales, but its reign was brief. In 2009, the country had fewer than four hundred thousand home LED bulbs installed; by 2014, it had seventy-eight million. In 2015, only about 4 percent of households used LEDs for most of their light; by 2020, that share had leapt to nearly half, while the CFL's share of homes fell from a third to about a tenth. The incandescent lit the country for a hundred years; the CFL peaked and collapsed inside of fifteen.

A change that large, arriving that fast, deserves more than a glance at the wattage. Each new light was tested for how many lumens it produced per watt. None was tested, before it filled the nation's bedrooms and nurseries, for what a lifetime of evenings under its spectrum does to a human being. The science that could answer that is only now catching up to the switch already made.

THE BASELINE: WHAT INCANDESCENT LIGHT ACTUALLY WAS

To judge the replacements you need a yardstick, and the incandescent bulb is it. Not out of nostalgia—out of physics. An incandescent bulb makes light the way the sun does and the way a fire does: by heating something until it glows. Run enough current through a tungsten filament

and it reaches thousands of degrees, and hot objects radiate light across the whole spectrum at once. Physicists call it blackbody radiation. The practical result is a smooth, continuous band of color, every wavelength present, none spiking above its neighbors.

That continuity is why colors look right under it. Lighting engineers score a light source on a color-rendering index, a scale that tops out at one hundred, where one hundred means colors look exactly as they do under natural daylight. A good incandescent bulb scores in the high nineties. Skin looks like skin, a tomato looks ripe, a bruise looks like a bruise. The bulb is not flattering the room; it is simply handing the eye a full deck of wavelengths to work with.

Two features of that spectrum matter most for the body. The first is where the energy sits. Incandescent light is weighted toward the warm end—heavy in red and orange, light in blue. The second is where it keeps going: the filament pours energy well past visible red into the near-infrared, past one thousand and beyond seventeen hundred nanometers, light the eye cannot see but the body still absorbs. Hold your hand near a lighted incandescent bulb and you feel that invisible tail as warmth. Keep both features in mind; each returns as something the newer bulbs took away.

The light of the incandescent bulb is close to the light the human body was built to read. For all but the last blink of human history, the only lights were the sun, the moon and fire—all warm, all continuous, all fading to red and dark at day's end. The eye and the internal clock learned their signals from that light. Where a bulb puts its energy is, therefore, not a decorator's question but a biological one. A landmark 2003 study measured the point: blue light near four hundred sixty nanometers suppressed the sleep hormone melatonin and shifted the body clock about twice as strongly as green light near five hundred fifty-five nanometers.¹ The lesson is not that blue light is poison—sunlight is full of it—but that a bulb's color balance carries a measurable biological signature, and blue at the wrong hour sends the wrong signal. Incandescent light, low in blue and rich in red, sits at the gentle end of that scale.

WHAT IS WRONG WITH THE COMPACT FLUORESCENT

The compact fluorescent lamp, the twisted-tube CFL, was the first mass replacement, pushed hard in the 2000s as the energy-saving bulb. It has two distinct problems: the light it makes, and the metal it needs to make it.

Start with the light. A fluorescent tube does not glow the way a filament does. It runs current through mercury vapor, which gives off ultraviolet light, which strikes a phosphor coating that glows in visible colors. That indirect route cannot produce a smooth curve. The CFL emits light in spikes—a few narrow bands the eye blends into something it reads as white. Color rendering suffers: a typical CFL scores in the low-to-mid eighties, well below the incandescent it replaced. Subtle colors flatten, skin takes on a grayish cast, and objects that reflect only the missing colors go dull. It is a coarse imitation of continuous light, assembled from a handful of bright lines.

Now the metal. Every CFL holds a small quantity of mercury, on the order of a few milligrams, sealed inside as vapor, because the tube cannot make light without it. Mercury is a recognized neurotoxin, and it is the reason a broken CFL is not treated like a broken drinking glass. What actually happens when one breaks in your home is a genuine scientific dispute, and it is worth reporting who says what, because the two sides weigh the evidence differently.

On the reassuring side stands a 2012 risk assessment by Nance and colleagues.² It modeled a CFL breaking in a room and compared the mercury readings against human toxicity benchmarks. For most scenarios, the hazard came out low—below the level of concern—though the authors noted that some cases, a bulb broken in a closed room with no cleanup did exceed their risk targets and warranted further study.² That paper is the one most often cited to show that broken CFLs are nothing to worry about.

Readers deserve to know where that reassurance came from. One of its authors, Michael Dourson, founded and directed a Cincinnati consulting organization called Toxicology Excellence for Risk Assessment (TERA).^{3,4} The Center for Public Integrity, in an investigation of the group, reported that more than half the

studies TERA reviewed were funded by the chemical industry.⁵ The American Chemical Society's own news outlet reported that at TERA, Dourson helped chemical and pesticide manufacturers evaluate their products' safety.⁶ The Washington Post reported that the group he founded produced research for chemical companies that consistently showed little or no human health risk from their products.⁷

The scrutiny reached a public peak in 2017. President Trump nominated Dourson to lead the Environmental Protection Agency's chemical safety office. The nomination collapsed in December of that year when it became clear the Senate would not confirm him, with Republican as well as Democratic senators citing his industry ties.⁶⁻⁹ TERA, for its part, describes itself as independent and says it does roughly two-thirds of its work for governments and nonprofits.^{3,4}

Two honest limits keep this from being an accusation. The Nance paper's own funding line names the New Zealand Ministry of Health and TERA; no evidence shows a bulb manufacturer paid for that particular study. The tie is to the author's organization and his broader work, as those outlets reported it—not to a line item on this paper. Weigh it for what it is, no more and no less.

On the concerned side stands independent work reaching a less comforting number. A 2012 European analysis found that mercury vapor from a CFL broken in a closed room could exceed the reference exposure limit set by the California Environmental Protection Agency.¹⁰ Even the reassuring study conceded that scenario. The disagreement is not about the chemistry but about how the risk should be framed: how often a real household breaks a bulb in a room it does not ventilate and does not clean up, whether a child is on the floor where vapor concentrates and how much margin a neurotoxin with no safe dose deserves. A reasonable person can read the same numbers and land on either side. The honest reading is that the question remains open.

There is a second mercury question the risk assessments rarely reach: what happens to the bulb after it dies. A CFL is not supposed to go in the household trash, because a landfill full of crushed tubes releases its mercury into soil and

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water, from which it returns to people through the food chain as methylmercury, the toxic form that accumulates in fish. Proper recycling exists, but surveys find that a large share of spent and broken CFLs are simply thrown away. The bulb marketed as the environmentally responsible choice depends, for that reputation, on a disposal discipline most households never practice.

Mercury is not the CFL's only health question. The tubes generate ultraviolet (UV) light as a step in making visible light, and the phosphor coating is supposed to trap it. When that coating is thin or cracked—and manufacturing leaves some bulbs with hairline cracks—a little UV escapes. A 2012 laboratory study from Stony Brook University exposed human skin cells to light from CFLs and found damage consistent with ultraviolet injury: reduced growth, signs of stress and effects strongest close to the bulb.¹¹ The finding is real but bounded: the study involved cells in a dish, not living people; skin has defenses a bare cell culture does not; and the effect fell off quickly with distance. Even so, health regulators in the United States and Canada advise against using single-envelope CFLs closer than about a foot from the skin—a modest precaution that itself concedes the emission is real.

WHAT IS WRONG WITH THE LED

The light-emitting diode replaced the CFL and now lights most of the country. It deserves credit first: an LED contains no mercury, so the breakage and disposal worries of the CFL disappear. It lasts far longer and uses even less power. On the two problems that plagued the fluorescent, the LED is a clear improvement. But it carries its own signature, and the signature is, in a specific way, the opposite of the filament's.

A white LED is not naturally white. It begins as a blue chip emitting near four hundred fifty nanometers, coated with a yellow phosphor that fills in enough of the other colors to fool the eye into seeing white. That manufacturing origin leaves three marks on the spectrum, and each one matters.

The first mark is a sharp blue spike. Right where the incandescent filament had a gentle rise, the LED has a tall narrow peak near four

hundred fifty nanometers—the raw output of the blue chip through the phosphor. The second mark is a dip in the cyan, around four hundred eighty to five hundred nanometers, between the blue spike and the phosphor's yellow hump. That valley matters more than its size suggests, because the eye's clock-setting cells peak in sensitivity right there, near four hundred eighty. The LED skips the middle of that region while spiking hard at four hundred fifty, which the same cells also read—stimulating the circadian system unevenly, strong where it disrupts, thin where the daytime signal sat. The third mark is a cliff at the red end. Even a warm twenty-seven-hundred-kelvin LED sold as cozy falls off near seven hundred nanometers and delivers almost nothing beyond. The near-infrared tail that poured out of the filament is simply gone, because the LED makes light by a different physics entirely.

The blue spike is the part the body reacts to at night. Building on the finding of the 2003 study mentioned earlier,¹ a 2022 study in the Proceedings of the National Academy of Sciences traced how the eye's photoreceptors combine to set the clock.¹² In the early minutes of exposure, the blue-sensitive cones carry much of the signal; as the exposure lengthens, a pigment called melanopsin, in a special class of retinal cells, takes over.¹² Blue-rich light reaches the clock through more than one channel, in other words, and it does so with a speed and a persistence that matter for anyone reading or watching a screen in the evening. Cool-white LED, the bright bluish kind common in kitchens and offices, is exactly the blue-rich light that pushes hardest on that system.

This is not a laboratory abstraction confined to monochromatic lamps and volunteers. In 2020, researchers at Monash University sent people home with a wearable light meter and measured the actual light in their houses at night. The result is the single most striking number in this whole subject. Homes lit with energy-efficient bulbs delivered nearly double the clock-relevant light compared to homes lit with incandescent bulbs, and nearly half of all the homes were bright enough to cut the occupant's melatonin roughly in half.¹³ The switch that was sold as saving energy also, quietly, in

an ordinary evening at home, roughly doubled the nightly dose of light the body reads as a signal to stay awake. Nobody chose that trade. It rode along with the bulb.

There is a fourth problem the spectrum charts do not show: flicker. Many LEDs, depending on the cheap driver inside them, rapidly pulse their brightness in a way the conscious eye does not register but the visual system still detects. Engineering standards exist because bad flicker is linked to headaches and eyestrain. A well-made LED flickers little; a bargain bulb can flicker badly, and nothing on the package says which is which.

Glare is a cousin of the same problem—an LED packs its light into a tiny, intense source, harsh in a way a frosted incandescent globe never was.

Honesty runs both ways here. On whether ordinary LED light physically injures the retina—the “blue light will blind you” claim that circulates online—the mainstream scientific bodies are reassuring, and this article will not pretend otherwise. The International Commission on Non-Ionizing Radiation Protection, an independent expert body, concluded in 2020 that typical exposure to blue or white LEDs has not been shown to cause acute damage to the retina.¹⁴

The clearest acute effect of LED lighting is not retinal injury but flicker, which lighting-engineering standards single out as its own hazard, separate from any question of photochemical harm. The real concern is not that the light burns the eye during normal use, but something subtler: what its nighttime rhythm does to the sleeping clock, and what its missing wavelengths quietly fail to do for the rest of the body. That second point is where the whole story turns.

One more light is coming, and it is worth knowing before it arrives. The brightest white source now emerging is the laser diode—a blue laser paired with a phosphor, the very same architecture as the white LED, only with a laser doing the emitting.¹⁵ It already lights the road: BMW and others have put laser headlights on high-end cars, and Nichia Corporation of Japan sells a white laser module for automotive and specialty use. It is not in home fixtures yet. But note the design: a laser’s output is narrower still than an LED’s, so the successor does not escape the blue-spike, red-cliff problem—it sharpens the starting point.

THE WAVELENGTH

YOU NO LONGER RECEIVE

The account so far is a ledger of harms added: a broken spectrum, a load of mercury, blue light at the wrong hour, flicker and glare. But the incandescent filament did something no ledger of harms can capture. It gave off a benefit—a steady, gentle bath of red and near-infrared light—and switching bulbs took that benefit away without anyone deciding to remove it or even noticing it was there. To understand what was lost, you have to know that red and near-infrared light is not biologically inert. The body does something with it.

What it does happens inside the cell, in the mitochondria—the power plants that turn food and oxygen into energy. Near the end of that chain sits an enzyme called cytochrome c oxidase, which absorbs red and near-infrared light. A widely cited 2016 review laid out the mechanism: when those wavelengths are absorbed, the enzyme works faster, energy production rises, the signaling molecule nitric oxide is released, and repair and anti-stress responses follow.¹⁶ A companion 2017 review traced the same light to reduced inflammation, with a biphasic dose-response: a modest amount helps, more can reverse the gain.¹⁷ This is the science now sold in red-light therapy panels. The incandescent bulb delivered a mild, unbranded version for free, every evening, as a side effect of being hot enough to glow.

The eye may register the loss most sharply, because the retina is one of the most energy-hungry tissues in the body and leans heavily on healthy mitochondria. In early 2026, researchers at University College London reported a striking

DO FULL-SPECTRUM LEDS EXIST?

Partly. A “high-CRI” or “full-spectrum” LED, of the kind sold by Waveform and a few other specialty makers, is a real step above the hardware-store LED in two ways. First, it fills in the visible spectrum more completely — a stronger red end and a softer blue spike, sometimes by starting from a violet chip rather than a blue one. Colors look closer to how they look under a filament, and the cyan dip described in the article is shallower. Second, a well-built driver flickers very little. Price does track driver quality, and the better makers publish their flicker figures, which most will not. If you must have LEDs, that is the kind to buy, in a warm color — 2700 kelvin or lower — for evening rooms.

Two limits remain, and no LED escapes them. No LED produces the near-infrared tail; that light comes from heat, and an LED is built not to make heat. And any white LED, however warm, still delivers more blue per unit of brightness at night than a filament does. So a full-spectrum LED is the best of the new bulbs, not a replacement for the old one. The honest label would be “fuller-spectrum.”

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result: supplementing narrow, LED-type light with a wider spectrum—the broad band that daylight and incandescent light contain—improved color-contrast sensitivity in people, and the improvement lasted about two months after only two weeks of exposure.¹⁸ The mechanism fits: short blue wavelengths near four hundred twenty to four hundred fifty nanometers suppress mitochondrial respiration in the retina; red wavelengths from about six hundred seventy upward increase it. Read those numbers against the LED spectrum. The LED is tall exactly where light suppresses retinal energy, and nearly empty exactly where light restores it. The incandescent filament was the reverse. The switch inverted the ratio the retina prefers.

It helps to picture the dose a person used to receive. For nearly all of human history, the end of the day was red—the setting sun reddening through the atmosphere, firelight and candlelight deep in the red and near-infrared. For the last century, the incandescent bulb carried that same warm, infrared-tailed light into the evening home. A person reading by a lamp after dark was bathed, mildly and continuously, in the wavelengths a cell uses to make energy and calm inflammation. Replace that lamp with a warm-white LED and the visible glow looks much the same, while the red and near-infrared behind it goes dark. The evening dose the body relied on for generations dropped, in a single household upgrade, close to zero.

The clinical picture deserves the same even hand the mercury question got. Much of the enthusiasm for red-light therapy of the aging eye traces to trials tied to the companies that make the devices. A 2024 overview in *Frontiers in Ophthalmology*, two of whose authors are affiliated with a light-therapy device maker, reads the mechanism generously.¹⁹ An independent 2024 meta-analysis weighed the pooled data more coldly: a real but small statistical benefit for dry age-related macular degeneration, and doubt whether it is large enough to matter to a patient's vision, given small samples and bias risk.²⁰ The mechanism is solid and the everyday benefit of red-rich light is plausible; the medical-grade claims remain contested. But strip the dispute to what nobody argues about, and one physical fact stands: the wavelengths that do this cellular

work were abundant in the bulb we retired and absent from the one that replaced it. We did not decide the benefit was worthless. We stopped receiving it without ever putting it on the ballot.

Step back and the shape of the loss is clear. The incandescent bulb was, by the measure of human biology, the best light in the room. It was also the least efficient, and efficiency is what the rules were written to reward. So, the one bulb whose light most resembled the light we evolved under is the one you can no longer easily buy new. That is a strange outcome for a country that prizes choice, and it did not happen by accident.

WHO OWNED THE LIGHT: A TALE OF THREE PATENTS

The bulbs did not leave the shelf on their own; a sequence of federal efficiency standards pushed them off, one rung at a time. The Energy Independence and Security Act of 2007 set the first minimums, and between 2012 and 2014 the traditional incandescent—one hundred watts first, then down to forty—could no longer be made or sold. A tougher backstop followed: a floor of forty-five lumens per watt, advanced in 2017, rolled back under the Trump administration, reinstated under the Biden administration in 2022, and enforced from August 2023. Incandescents make about fifteen lumens per watt and halogens twenty-five, so that rule ended them outright. (Halogen bulbs are a type of incandescent bulb, only hotter.)

A further standard, proposed in 2022 and finalized in 2024, of more than one hundred twenty lumens per watt takes effect in July 2028 and ends the compact fluorescent.²¹ The Department of Energy frames each step as saving consumers money and energy, and on its own terms it does. Notice, though, what an efficiency rule must do. It falls hardest on the least efficient bulb—always the oldest one, because efficiency climbs with each generation. The oldest technology is also, by the same passage of time, the one whose patents expired longest ago. So, the design pushed off the shelf is the one already in the public domain, and the design left standing is the newer, patented one. No one need aim the rule at the patents for the two to line up. Age lines them up.

Behind each of these bulbs sits a patent, and behind each patent sits an owner. Line the three up by age and a pattern appears—a pattern, not a plot.

The incandescent bulb traces to Edison's patents of the 1880s. They expired well over a century ago. The design has belonged to the public for longer than anyone alive can remember—and it is the one now gone from the shelf.

The spiral compact fluorescent was a General Electric (GE) invention. GE's engineer, Edward Hammer, bent a fluorescent tube into a spiral in 1976 during the energy crisis, and the company patented the work. Those patents expired long ago as well; the CFL, too, is public-domain—and next in line to go.

The white LED is the newcomer, and its story is the sharpest. It exists because of one company: Nichia, a phosphor maker whose engineer Shuji Nakamura built the first efficient blue LED in the early 1990s. In 1996, Nichia paired a blue chip with a yellow phosphor and produced the first commercial white LED—the very design lighting your ceiling now. The work won a share of the 2014 Nobel Prize in Physics. Nichia held the patents, earned a reported fortune from them and defended them hard, once suing the American manufacturer Cree.^{22,23} The white light in your home began as one firm's closely held property, and its core patents, filed in the 1990s, have only recently begun to expire.

This article alleges no scheme and proves no motive; the record here does not show who wanted what. It shows a correlation, and the correlation carries a plain logic.

A design in the public domain earns no one a premium. Anyone may build it; competition drives the price toward cost and the profit toward zero. A patented design is different: for the twenty years of its term, one owner may charge what the monopoly allows. So, the money in lighting

always sits with whatever is newest and still protected, never with the old design the public owns. That is not a claim about anyone's intentions. It is simply where the incentive points.

The pattern has a next chapter already forming. As the LED's patents lapse, the same inventor behind the blue LED, Shuji Nakamura, and the same company, Nichia, have each separately moved to laser-based white light—a fresh patent portfolio for the newest source, arriving just as the technology it replaces slips into the public domain. Right on schedule.¹⁵

Notice what this cadence does to knowledge. Sound science on a technology's long-term health effects is rarely settled in under twenty years—you cannot measure a twenty-year effect in less time, and consensus usually takes longer still. Tobacco took decades; leaded gasoline took generations. But each lighting technology is pushed aside by its patented successor on roughly that same clock. The verdict on what a bulb does to the body arrives about when the bulb is already being retired for the next one—whose own effects no one has yet had time to study. The evidence never catches up.

THE LESSER OF BAD CHOICES

Stand in the lighting aisle today and the choice is worse than it looks. The incandescent,

WHY YOU CAN'T JUST BUY THE OLD BULB

The healthy bulb is missing because of a rule, not a danger—and here is how that rule works. The Constitution lets Congress grant a patent—an exclusive right to make and sell an invention—but only for “limited Times.” That is a bargain: the inventor gets a temporary monopoly, and in exchange the public gets the invention forever once the clock runs out. The design enters the public domain, free for anyone to build. The incandescent's patents expired long ago. By that bargain, the bulb belongs to all of us now. Nobody owns it.

Here is the collision. Federal energy-efficiency regulation sets a minimum on how much light a bulb must produce per watt. The ordinary incandescent cannot meet it, because turning most of its energy into heat is how a glowing filament works. The rule does not call the design dangerous or defective. It says the design may not be sold—a design the public already owns, that no patent covers, blocked at the point of sale.

The effect is a kind of second monopoly the Constitution never authorized. The patent monopoly was supposed to end, and the public was supposed to inherit the freedom to build. Instead, a regulatory wall rose in its place. The “limited Time” became, in practice, unlimited—not by extending the patent, which would be plainly illegal, but by forbidding the sale of what the expired patent set free.

Who can fix it? Not, realistically, the legislature that built the wall. The more promising path runs through the courts, which can be asked a narrow question: when regulation permanently blocks the manufacture of a design whose patent has expired, does that defeat the Constitution's guarantee of a public domain after “limited Times”? The wall was built by human decision. It can be removed by one.

None of this asks you to believe the old bulb is harmless or that efficiency does not matter. It asks something smaller: that a design the public already owns should not be forbidden to the public without a far better reason than the electric bill—especially now that the light we were pushed toward carries costs to the body the light we lost did not. The full argument, with its Supreme Court cases and figures, is laid out in my companion web article.²⁴

spectrally closest to sunlight, is largely gone from the shelf. The CFL carries mercury and a broken spectrum. The LED solves the mercury but strips out the red and near-infrared the body used, spikes the blue the clock reacts to at night, and may flicker in ways the package never mentions. Every option in front of you is the lesser of several bad choices, and the best of them by the measure of human biology is the one that was regulated away.

Why can you not simply buy the old bulb? The answer is not chemistry or physics. It is law—and it turns on a phrase in the Constitution most Americans have never read. The Patent Clause gives an inventor an exclusive right for “limited Times,” after which the design is meant to enter the public domain, free for anyone to build. A patent-expired incandescent bulb should, by that promise, be free to make and sell forever. Federal energy regulation stands in the way, and the sidebar explains how that collision works and why the courts, not the legislature, may set it right. The full argument lives at the companion web article.²⁴ For now, it is enough to notice the health cost hiding inside an energy policy: in the name of the electric bill, we traded away the light our biology was tuned to—the light your grandmother bought without a second thought.

And here the two halves of this story turn out to be one. The healthiest light is not missing because science failed to invent it or because the market would not sell it. It is missing because a regulation withdrew a design the public already owned—and that is not a health problem with a legal footnote, it is a health problem made of law. No diet, no supplement, no red-light gadget reaches the cause, because the cause is a rule. The market cannot repeal it; the agency that wrote it will not; the legislature that authorized it is least likely of all to. One lever remains. The Constitution promised that an expired patent returns its design to the public, free to build again—the “limited Times” the Framers wrote down. A court can hold the government to that promise. That is why the health of the light and the fate of the Patent Clause cannot be pulled apart: the clause is the only door back to the bulb your body was built for.

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